Working of master cylinder

When the brake pedal is pressed

Normally, the piston inside the master cylinder is positioned in the backward position due to the action of the tensile force of the return spring. Hence due to the opening of inlet and bypass ports the brake oil is filled at the front and back side of the piston. When the brake pedal is pressed by the driver, the piston rod connected to the brake pedal pushes the piston in the forward direction. Because of this action, the two holes are sealed and fluid is pressed well. The compressed fluid is then passed through the one way valve to the wheel cylinder through the pipe lines connected with it. Now the two pistons inside the wheel cylinder are pushed outward. Because of this action the brake shoes are expanded with help of the piston's push rod which forces the brake drum to stop from rotation and stops the wheel from the rotation.

When the brake pedal is released

The pressure inside the master cylinder is reduced. At the same time due to the tensile force of return spring brake shoes are relieved from the brake drum and pulled towards in. Hence the pistons inside the wheel cylinders are moved inward. Hence the brake fluid which is staying in the pipes is forced backwards and sent to the master cylinder through the test valve.

In the master cylinder when the piston moving in backward direction, the brake fluid is initially sent to the reservoir through bypass port. At the same time through the feed holes in the piston the brake fluid is moved to the front side of the piston. Hence the required fluid for the next piston operation is made available at the front side of piston.

Tandem master cylinder

In normal master cylinders only one outlet is generally provided. Moreover, from the location point of view, the master cylinder is located very near to the front wheels and the rear wheels are far away from the master cylinders. Hence, while braking the time taken by the brake fluid to travel from the master cylinder to reach the rear wheel cylinder becomes long. Hence the applied braking force is varied for both the front and rear wheels. Hence, Tandem master cylinder was designed to overcome the above said problem. The photographic view of the tandem master cylinder can be seen in Figure.

In this cylinder there are separate outlets available for front and rear wheels. The brake fluid is passed to the wheel cylinders through separate pipelines and the braking action is performed. In this type of master cylinders, initially the brake fluid is sent to the rear wheels and then to the front wheels. By that way equal amount of braking force is applied at a time for both the wheels. Hence the chance for the faulty operation or faulty situation during braking is eliminated. In the modern vehicles mostly tandem type master cylinder is used. The construction and working principle of the Tandem master cylinder can be seen in Figure.

While pressing brake pedal

Similar to the normal master cylinder, tandem master cylinder also has pistons, primary rubber cap, return

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